

LEEDS WEATHER CENTRE, United Kingdom

WMO: 033470

Lat: 53.800N

Lon: 1.550W

Elev: 154

StdP: 14.61

Time Zone: 0.00 (EUW)

Period: 94-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	28.2	30.1	20.7	15.6	35.2	23.1	17.5	35.0	34.6	45.6	28.7	45.5	5.5	300	0.585

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.3	79.8	64.7	75.8	62.8	72.7	61.2	66.0	75.9	64.3	73.6	62.8	70.6	8.5	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.4	84.7	70.0	60.8	80.1	67.9	59.3	75.8	66.1	30.8	75.8	29.5	73.7	28.4	70.8	71.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.0	23.7	20.3	DB	23.7	85.4	3.2	2.8	21.4	87.4	19.5	89.0	17.7	90.6	15.4	92.7	(4)
(5)			WB	22.9	68.6	3.1	1.4	20.7	69.6	18.8	70.4	17.1	71.2	14.9	72.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.9	40.9	42.2	44.6	47.9	53.5	58.4	62.8	63.1	57.4	51.9	46.0	40.8
	DBStd	9.43	5.12	5.44	5.10	5.58	5.60	4.87	4.54	4.95	3.94	5.22	5.13	5.83
	HDD50	1288	282	224	177	107	28	1	0	0	1	40	140	288
	HDD65	5249	746	639	632	514	356	207	100	98	231	406	571	749
	CDD50	1599	1	5	10	43	138	252	397	407	223	100	19	4
	CDD65	84	0	0	0	0	1	8	31	40	4	0	0	0
	CDH74	519	0	0	0	1	15	63	173	245	22	0	0	0
	CDH80	94	0	0	0	0	0	14	21	58	1	0	0	0
Wind	WSAvg	8.7	9.4	11.4	10.3	8.6	8.1	8.9	8.4	7.5	7.5	8.4	8.0	8.4
Precipitation	PrecAvg	28.9	2.6	1.9	2.3	2.2	2.4	2.2	2.0	2.8	2.6	2.5	2.8	2.7
	PrecMax	36.1	5.9	6.4	4.4	5.0	5.8	11.9	4.7	5.2	8.5	6.1	5.4	7.6
	PrecMin	19.4	0.8	0.2	0.3	0.3	0.3	0.4	0.4	0.8	0.4	0.5	1.0	1.1
	PrecStd	4.7	1.3	1.3	1.1	1.3	1.3	2.2	1.0	1.1	1.9	1.4	1.3	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.7	58.6	60.3	68.4	76.4	81.5	83.2	85.7	77.7	68.6	59.6	57.6
		MCWB	50.8	52.1	49.8	53.9	60.6	66.2	63.8	65.5	65.3	59.3	55.3	53.6
	2%	DB	52.5	54.6	57.4	63.1	71.6	74.1	78.6	80.8	70.1	64.0	57.5	54.9
		MCWB	48.6	49.9	49.7	52.5	58.5	60.5	63.8	65.3	60.7	57.2	53.5	51.2
	5%	DB	50.7	52.1	54.8	59.9	66.7	70.6	75.3	75.6	66.7	61.7	55.4	51.3
		MCWB	47.0	47.7	48.7	50.9	56.3	59.1	62.5	63.5	58.4	56.0	51.7	47.8
	10%	DB	49.0	50.5	52.8	57.1	63.3	67.5	72.2	71.9	64.2	59.7	53.5	49.0
		MCWB	45.7	46.2	47.3	49.5	54.4	57.5	61.3	61.8	57.0	54.9	50.3	46.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.3	52.3	53.4	56.4	62.8	66.2	67.5	68.5	66.2	60.5	56.3	54.0
		MCDB	54.6	56.7	57.6	65.2	72.7	78.0	77.1	79.4	73.5	65.0	58.7	57.2
	2%	WB	49.2	50.4	51.3	54.0	59.5	62.6	65.5	66.2	62.7	58.7	54.3	51.7
		MCDB	52.1	54.2	55.5	61.1	68.2	71.6	74.8	76.5	69.0	62.8	57.0	54.7
	5%	WB	47.6	48.4	49.4	51.9	57.4	60.4	63.9	64.6	60.2	57.1	52.3	48.5
		MCDB	50.3	51.8	53.5	58.4	65.6	67.8	72.5	74.3	64.2	60.6	54.6	50.6
	10%	WB	46.0	46.5	47.9	50.2	55.3	58.6	62.3	63.1	58.5	55.5	50.8	46.4
		MCDB	48.7	50.1	52.3	56.0	62.2	65.7	70.2	71.0	62.3	59.2	53.0	48.9
Mean Daily Temperature Range	5% DB	MDBR	7.8	9.8	11.3	12.9	14.0	13.6	14.2	14.3	12.1	10.9	8.2	7.7
		MCDBR	10.0	10.8	14.3	16.6	19.8	20.2	20.0	20.9	17.0	12.1	9.2	9.5
		MCWBR	8.6	8.9	9.0	8.4	10.2	9.8	8.3	8.3	9.0	7.9	7.6	8.1
	5% WB	MCDBR	9.7	10.5	12.0	15.2	19.3	17.4	17.5	17.8	13.8	11.8	8.7	9.4
		MCWBR	8.7	9.3	9.0	8.5	11.0	9.9	8.2	7.9	8.9	9.0	7.9	8.7
Clear-Sky Solar Irradiance	taub	0.322	0.337	0.367	0.395	0.395	0.398	0.398	0.385	0.366	0.348	0.322	0.310	
	taud	2.325	2.374	2.293	2.236	2.275	2.294	2.296	2.351	2.412	2.435	2.388	2.305	
	Ebn at Noon	201	240	256	264	270	270	267	265	255	233	201	179	
	Edh at Noon	19	24	32	39	39	39	38	34	29	23	17	16	
All-Sky Solar Radiation	RadAvg	204	414	757	1181	1494	1536	1490	1227	897	507	251	161	
	RadStd	22	47	92	137	95	141	148	80	81	44	27	16	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (66 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page